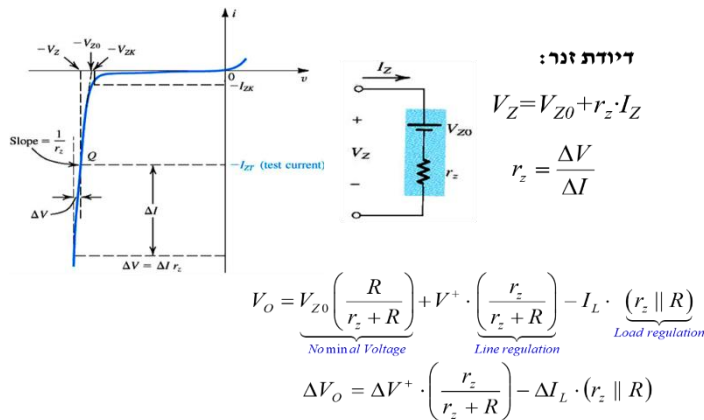


נוסחות קורס אלקטרוניקה תקבילית

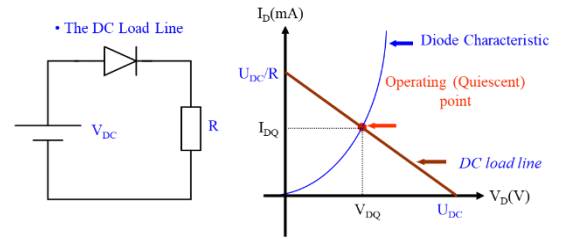
דיודה:



$$I_D = I_S \cdot e^{\frac{V_D}{nV_T}} \Leftrightarrow V_D = nV_T \ln(I_D/I_S)$$

$$V_T = \frac{kT}{q}; I_S(T) = I_S(T_0) \times 2^{\frac{T-T_0}{5}}$$

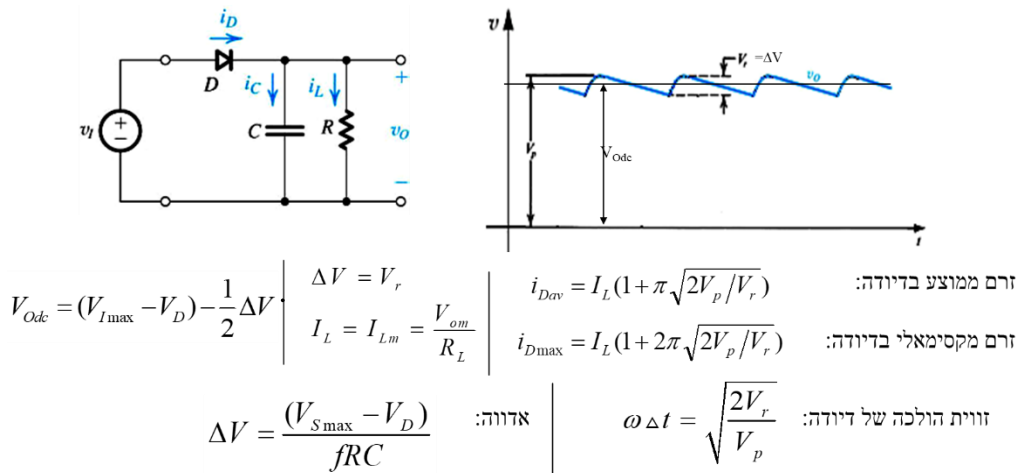
$$V_{D1} - V_{D2} = nV_T \ln(I_{D1}/I_{D2}) = 2.3 \cdot n \cdot V_T \log(I_{D1}/I_{D2})$$



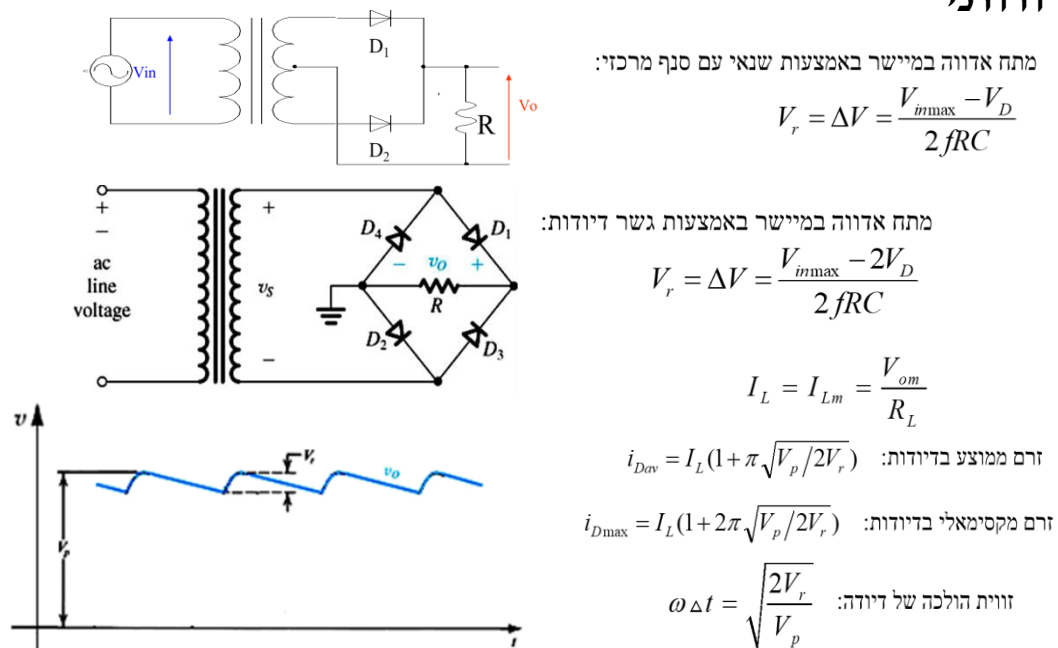
$$r_D = \left. \frac{dV_D}{dI_D} \right|_{Q\text{-point}} = \frac{nV_T}{I_{DQ}} = \frac{n \times 25mV}{I_{DQ}}$$

דיודה לאות קטן:
(צומת NP-PN)

מיישר חד דרכי



מיישר דו דרכי



$$v_{gs} \leq 0.2 \cdot (V_{GSQ} - V_t) = 0.2 V_{OVQ}$$

$$r_0 = \frac{V_A}{I_{DQ}} = \frac{1}{\lambda I_{DQ}}$$

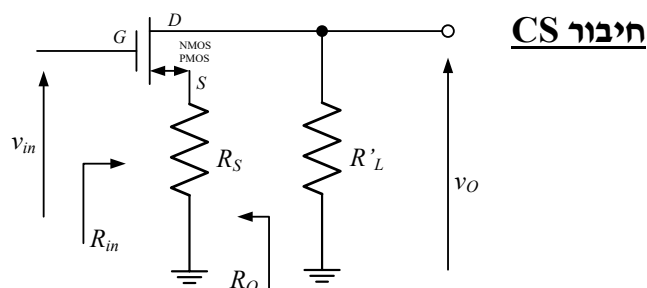
$$g_m = k'_n \left(\frac{W}{L} \right) \cdot (V_{GSQ} - V_t) = k'_n \left(\frac{W}{L} \right) V_{OVQ}$$

$$g_m = \sqrt{2k'_n \cdot (W/L) \cdot I_{DQ}}$$

$$g_m = \frac{2I_{DQ}}{V_{GSQ} - V_t} = \frac{2I_{DQ}}{V_{OVQ}}$$

$$V_{Omax}|_{v_O=v_{ds}} = \min \left[(V_{DSQ} - V_{GSQ} + V_t), I_{DQ} R_{AC} \right]$$

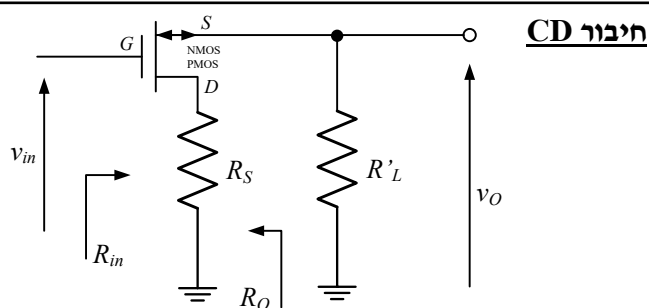
$$\begin{cases} V_{Omax}|_{v_O \neq v_{ds}} = I_{Dmax} R'_L \\ I_{Dmax} = \min \left[\left(\frac{V_{DSQ} - V_{GSQ} + V_t}{R_{AC}} \right), I_{DQ} \right] \end{cases}$$



$$R_{in} \rightarrow \infty$$

$$A_v = \frac{v_o}{v_i} = - \frac{g_m R'_L}{(1 + g_m R_S) + \frac{R_S + R'_L}{r_o}} \bigg|_{r_o \rightarrow \infty} = - \frac{g_m R'_L}{(1 + g_m R_S)}$$

$$R_o = (g_m R_S + 1) r_o + R_S \big|_{r_o \rightarrow \infty} = \infty$$



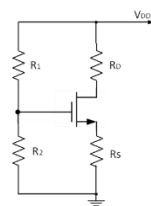
$$R_{in} \rightarrow \infty$$

$$A_v \equiv \frac{v_o}{v_i} = \frac{g_m R_L \parallel r_o}{g_m (R_L \parallel r_o) + 1} \bigg|_{r_o \rightarrow \infty} = \frac{g_m R'_L}{1 + g_m R'_L}$$

$$R_{out} = R_o = r_o \parallel 1/g_m \big|_{r_o \rightarrow \infty} = \frac{1}{g_m}$$

MOSFET טרנזיסטור

ניתוח DC:



NMOS

$V_{GS} \leq V_t \Rightarrow$ נתק
 $V_{GS} > V_t$ הולכה

$V_t > 0$ Enhancement
 $V_t < 0$ Depletion

$$\begin{aligned} k'_n &= \mu_n C_{ox} \left[\frac{A}{V^2} \right] \\ K'_n &= \frac{1}{2} k'_n \left(\frac{W}{L} \right) \end{aligned}$$

רוויה:

$$I_D = \frac{1}{2} k'_n \left(\frac{W}{L} \right) (V_{GS} - V_t)^2$$

ללא אפקט התקצרות התעלה:

$$I_D = \frac{1}{2} k'_n \left(\frac{W}{L} \right) \cdot V_{OV}^2 (1 + \lambda V_{DS}); \lambda = \frac{1}{V_A}$$

עם אפקט התקצרות התעלה:

$$V_{DS} > V_{GS} - V_t = V_{OV}$$

$$R_{Dmax} = \frac{V_{DD} - V_{GSQ} + V_t}{I_{DQ}} - R_S$$

PMOS

$$V_{GS} \geq V_t \Rightarrow \text{נתק (1)}$$

$$V_{GS} < V_t \Rightarrow \text{הולכה (2)}$$

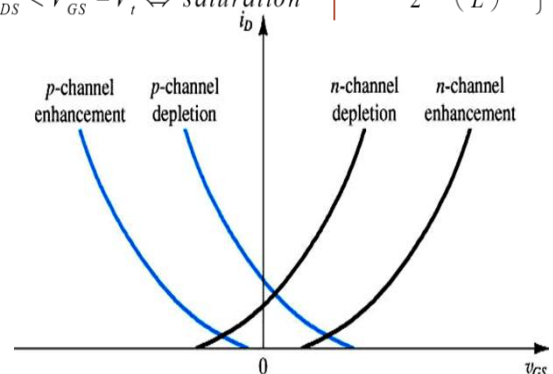
$V_t < 0$ Enhancement
 $V_t > 0$ Depletion

$$\begin{aligned} k'_p &= \mu_p C_{ox} \left[\frac{A}{V^2} \right] \\ K'_p &= \frac{1}{2} k'_p \left(\frac{W}{L} \right) \end{aligned}$$

נתון יצרן:

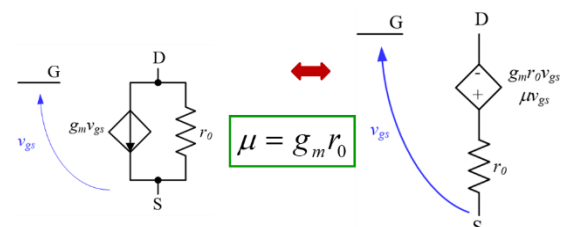
$$I_D = \frac{1}{2} k'_p \left(\frac{W}{L} \right) (V_{GS} - V_t)^2$$

$$V_{DS} < V_{GS} - V_t \Leftrightarrow \text{saturation}$$

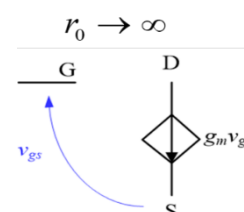


ניתוח AC:

מודל לאות קטן כולל אפקט התקצרות התעלה



מודל לאות קטן ללא אפקט התקצרות התעלה

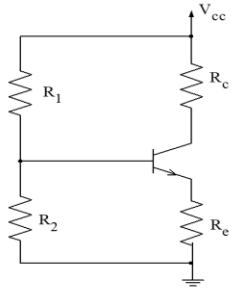
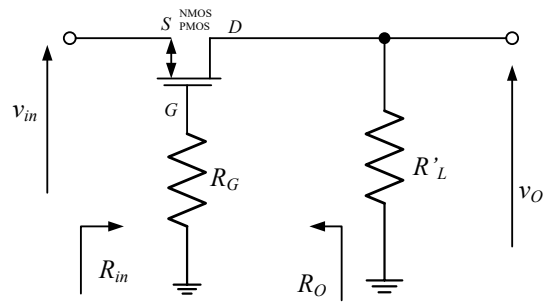


$$R'_L = R_D \parallel R_L$$

$$R_{in} = \frac{v_{gs}}{i_d} = \frac{r_o + R_D \parallel R_L}{1 + g_m r_o} = \frac{r_o + R'_L}{1 + g_m r_o} \Big|_{r_o \rightarrow \infty} = \frac{1}{g_m}$$

$$A_V = \frac{v_o}{v_{in}} = \frac{(1 + g_m r_o) R'_L}{r_o + R'_L} \Big|_{r_o \rightarrow \infty} = g_m R'_L$$

$$R_o = (g_m r_o + 1) R_{sig} \parallel R_S + r_o \Big|_{r_o \rightarrow \infty} \rightarrow \infty$$



$$V_{bb} = V_{TH} = \frac{V_{cc} R_2}{R_1 + R_2}$$

$$R_b = R_{TH} = R_1 \parallel R_2 = \frac{R_1 R_2}{R_1 + R_2}$$

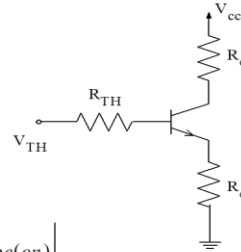
$$\text{assumption: } I_c = \beta I_b$$

$$\text{proof: } |V_{cc}| > |V_{ce} = V_{be} + V_{cb}| > |V_{be(on)}|$$

$$I_{bQ} = \frac{V_{bb} - V_{be}}{R_b + (\beta + 1) R_e}$$

$$I_{cQ} = \beta I_{bQ} = \frac{\beta (V_{bb} - V_{be})}{R_b + R_e (\beta + 1)}$$

$$V_{ceQ} = V_{cc} - I_c (R_c + R_e)$$



טרנזיסטור BJT

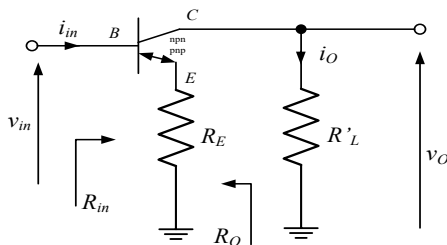
ניתוח DC:

$$\beta = \frac{\alpha}{1 - \alpha}; \alpha = \frac{\beta}{\beta + 1}$$

$$\text{if } v_O = v_{CE} \Rightarrow v_{Omax} = \min((V_{CEQ} - V_{CEsat}), (I_{CQ} R_{AC}))$$

$$\text{if } v_O \neq v_{CE} \Rightarrow i_{Cmax} = \min\left(I_{CQ}, \left(\frac{V_{CEQ} - V_{CEsat}}{R_{AC}}\right)\right); v_{Omax} = i_{Cmax} R'_L$$

חיבור CE



$$\frac{i_O}{i_i} = -\beta$$

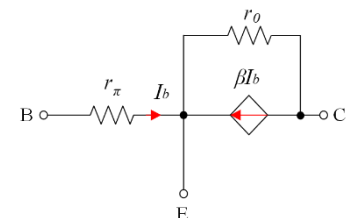
$$\frac{v_O}{v_i} = -\frac{\beta R'_L}{r_\pi + (\beta + 1) R_E} \Big|_{R_E=0} = -\frac{\beta R'_L}{r_\pi} = -g_m R'_L$$

$$R_{in} = r_\pi + R_E (\beta + 1) = (\beta + 1)(r_e + R_E)$$

$$R_O = \infty$$

ניתוח AC:

מודל לאות קטן כולל Early Effect:

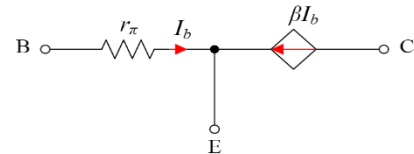


$$r_\pi = \frac{v_{be}}{i_b} = \frac{V_T}{I_B} = \frac{\beta V_T}{I_C}$$

$$r_e = \frac{r_\pi}{\beta + 1} \approx \frac{r_\pi}{\beta} = \frac{1}{g_m}$$

$$r_o = \frac{V_A}{I_C}$$

מודל לאות קטן ללא Early Effect:

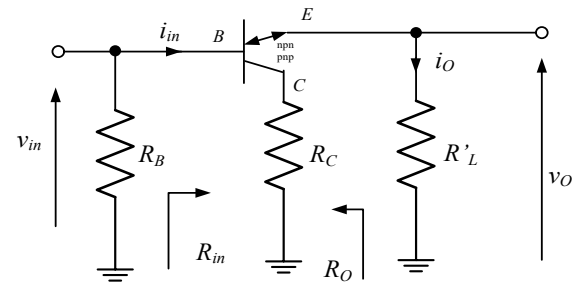


חיבור CC

$$R_{in} = r_{\pi} + (\beta + 1)R'_L = (\beta + 1)(r_e + R'_L)$$

$$\frac{i_o}{i_b} = \beta + 1 ; \quad \frac{v_o}{v_i} = \frac{(\beta + 1)R'_L}{r_{\pi} + (\beta + 1)R'_L} = \frac{R'_L}{R'_L + r_e}$$

$$R_o = \frac{R_B + r_{\pi}}{(\beta + 1)} = \frac{R_B}{(\beta + 1)} + r_e$$



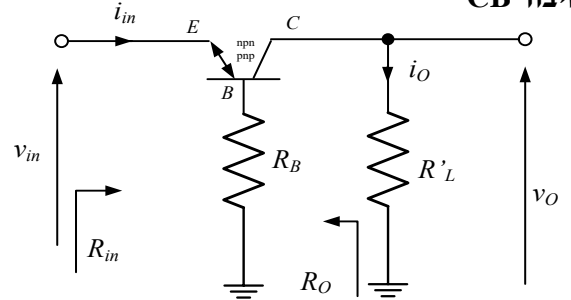
חיבור CB

$$R_{in} = \frac{r_{\pi} + R_B}{\beta + 1} \Big|_{R_B=0} = \frac{r_{\pi}}{\beta + 1} = r_e = \frac{1}{g_m}$$

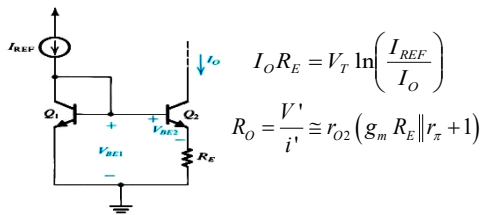
$$\frac{i_o}{i_i} = \alpha = \frac{\beta}{\beta + 1}$$

$$\frac{v_o}{v_i} = \frac{\beta R'_L}{r_{\pi} + R_B} \Big|_{R_B=0} = \frac{\beta R'_L}{r_{\pi}} = g_m R'_L$$

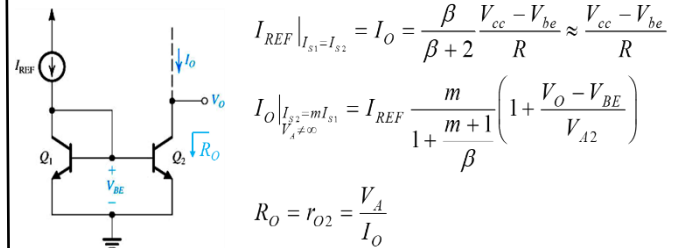
$$R_o = \infty$$



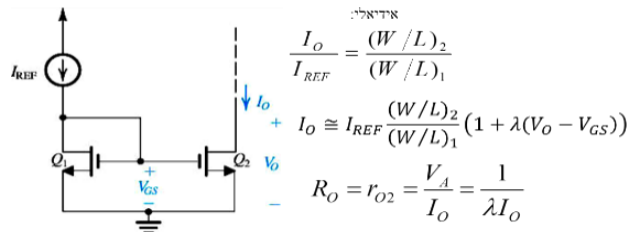
מקור זרם לזרמים של מיקרואמפרים ועשרות מיקרואמפרים



מקור זרם ראוי-BJT:



מקור זרם ראוי-FET:



מגבר הפרש BJT:

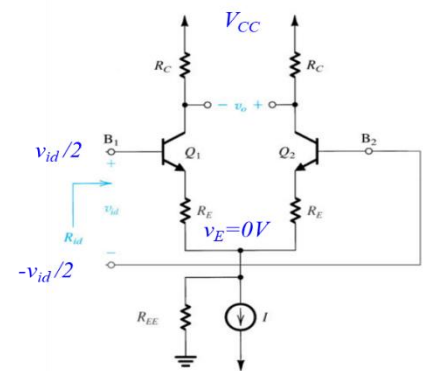
$$A_d = - \frac{\beta R_C r_0}{(r_{\pi} + R_E)(r_0 + R_C + R_E) + \beta R_E r_0}$$

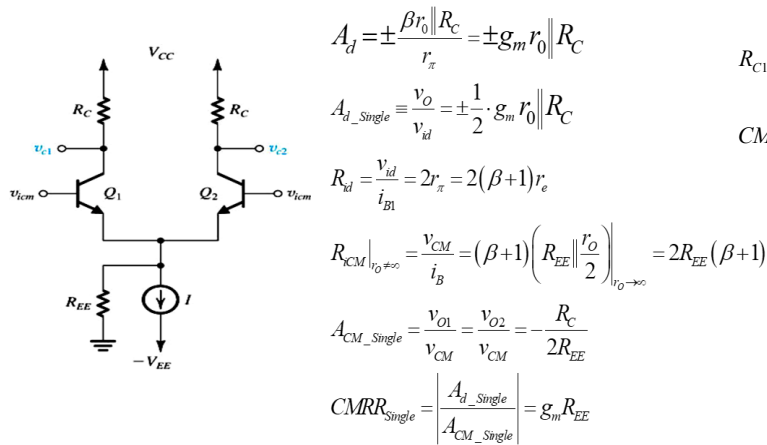
$$R_{id} = 2 \left((r_{\pi} + R_E) + \frac{R_E \beta r_0}{r_0 + R_C + R_E} \right)$$

$$A_d \Big|_{r_0 \rightarrow \infty} = - \frac{g_m R_C}{1 + \frac{R_E}{r_e}} = - \alpha \frac{R_C}{r_e + R_E}$$

$$R_{id} \Big|_{r_0 \rightarrow \infty} = 2(r_{\pi} + (\beta + 1)R_E)$$

$$A_{d_Single_ended} \equiv \frac{v_{O2}}{v_{id}} = - \frac{v_{O1}}{v_{id}} = \frac{1}{2} \cdot \frac{g_m R_C}{1 + \frac{R_E}{r_e}} = \frac{1}{2} \cdot \alpha \frac{R_C}{r_e + R_E}$$





$$R_{C1} \neq R_{C2} \Rightarrow A_{cm_diff} \cong \frac{\Delta R_C}{2R_{EE}} = A_{cm_single} \cdot \left(\frac{\Delta R_C}{R_C} \right)$$

$$CMRR_{diff} = \left| \frac{A_{d_diff}}{A_{cm_diff}} \right| = \frac{g_m R_C}{(\Delta R_C / 2R_{EE})} \cong 2g_m R_{EE} \left(\frac{R_C}{\Delta R_C} \right)$$

מגבר הפרש FET

$$0 < v_{id} \leq v_{id\max} = \sqrt{2}V_{OV} ; V_{CM\min} \leq V_{CM} \leq V_{CM\max}$$

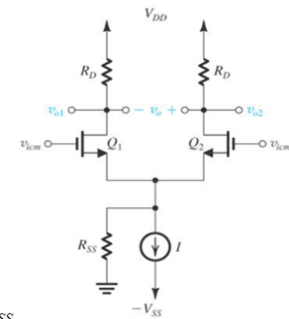
$$V_{CM\min} = V_{SS} + V_{CS\min} + V_{GSQ} ; V_{CM\max} = V_{DD} - \frac{IR_D}{2} + V_t$$

$$g_m = \frac{2I_D}{V_{GS} - V_t} = \frac{2I_D}{V_{OV}} = \frac{I}{V_{OV}}$$

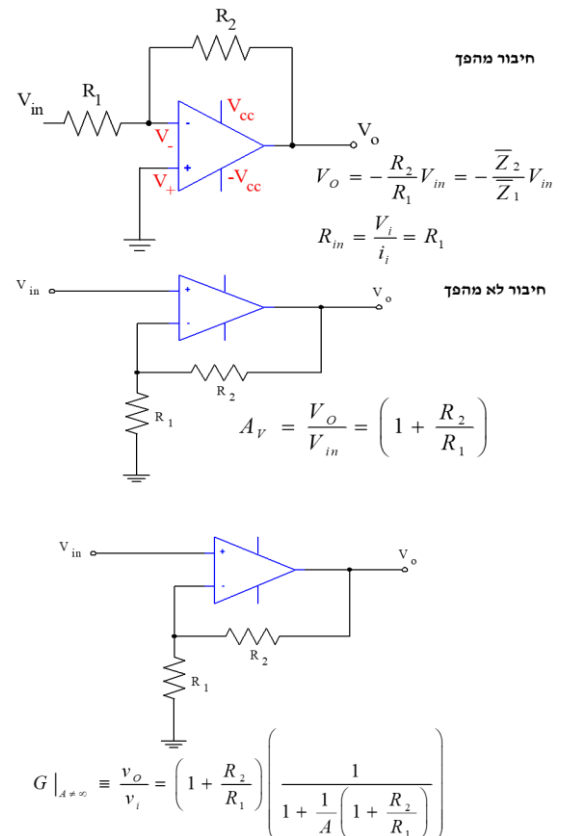
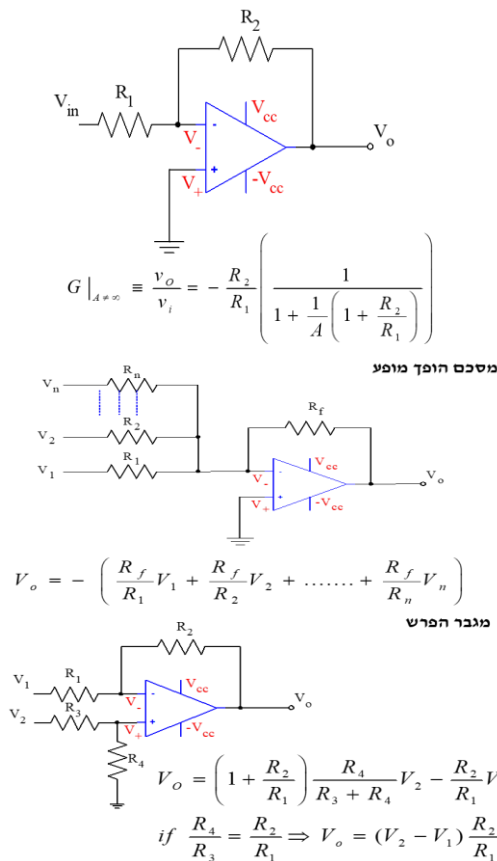
$$A_d = -g_m r_0 \| R_D ; A_{d_Single_ended} = \pm \frac{1}{2} g_m r_0 \| R_D$$

$$A_{CM_Single_ended} = \frac{v_{O2}}{v_{CM}} = \frac{v_{O1}}{v_{CM}} = -\frac{R_D}{2R_{SS}} ; CMRR_{Single_ended} = g_m R_{SS}$$

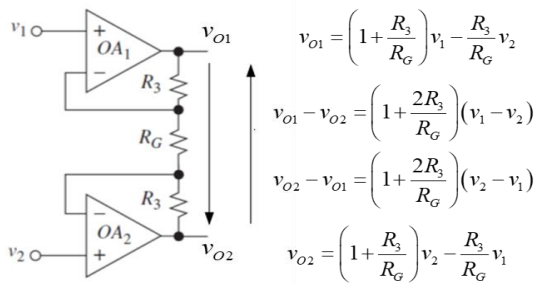
$$A_{CM_Single_ended} \Big|_{g_{m1} \neq g_{m2}} = \frac{R_D}{2R_{SS}} \times \frac{\Delta g_m}{g_m} ; CMRR_{Single_ended} \Big|_{g_{m1} \neq g_{m2}} = \frac{2g_m^2 R_{SS}}{\Delta g_m}$$



מגברי שרת אידאליים



מגבר מכשור



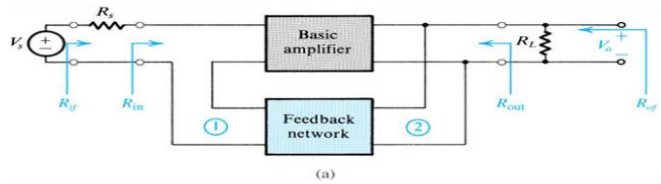
Slew Rate

$$V \cdot \omega_i \leq SR, \quad \omega_M V_{Omax} = SR$$

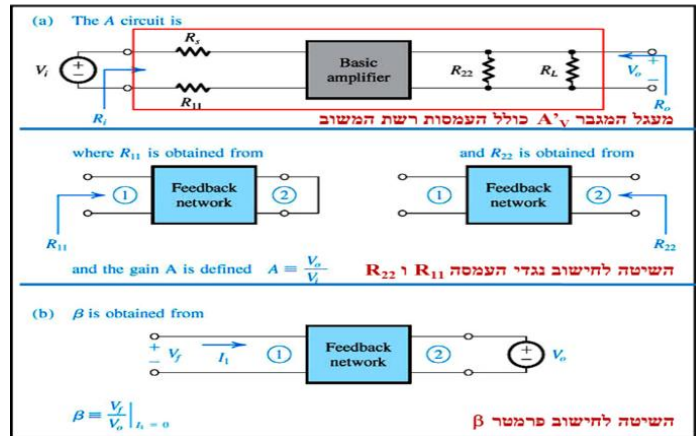
$$f_M - \text{full power bandwidth}$$

$$f_M = \frac{SR}{2\pi V_{Omax}} \Leftrightarrow V_O \omega = \omega_M V_{Omax}$$

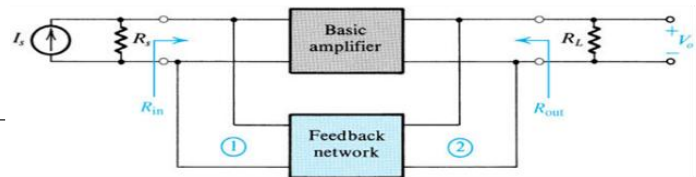
השוואת מתח – דגימת מתח (מתח, מתח)



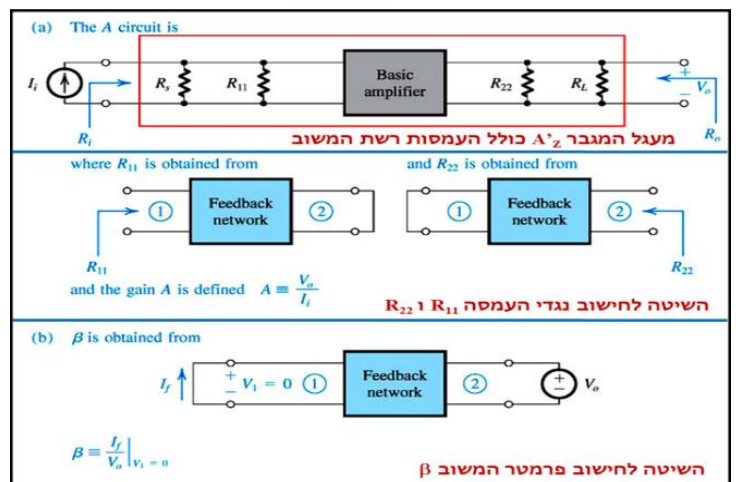
$$D = 1 + \beta A'_V, \quad A'_{Vf} = \frac{A'_V}{D}, \quad R_{if} = R_{in} D, \quad R_{of} = \frac{R_O}{D}$$



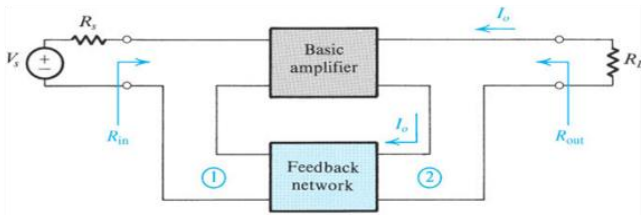
השוואת זרם – דגימת מתח (זרם, מתח)



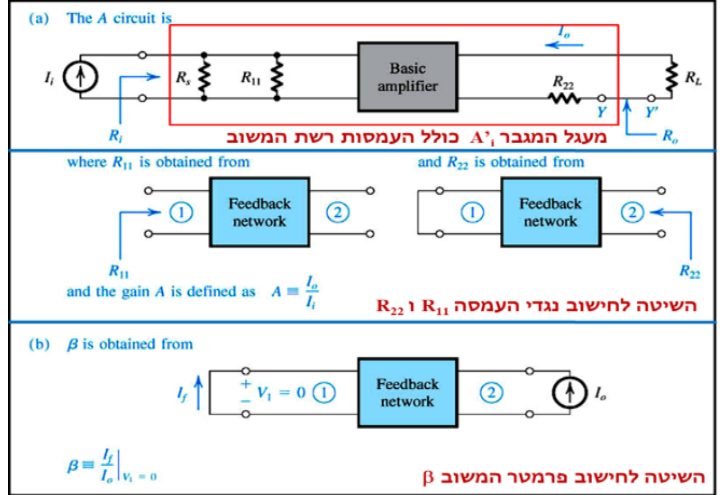
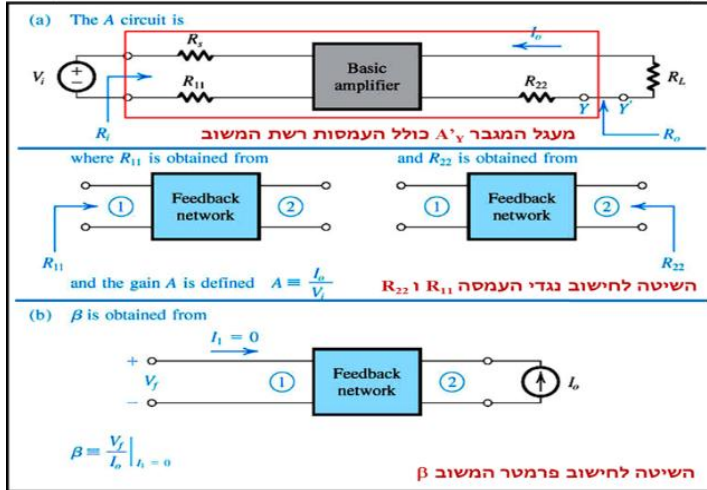
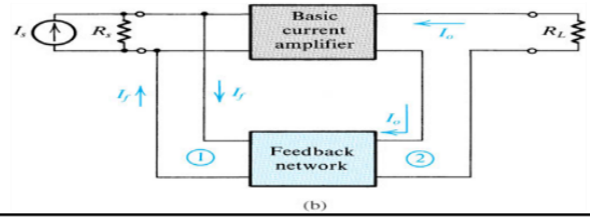
$$D = 1 + \beta A'_Z, \quad A'_{Zf} = \frac{A'_Z}{D}, \quad R_{if} = \frac{R_{in}}{D}, \quad R_{of} = \frac{R_O}{D}$$



השוואת מתח - דגימת זרם (מתח, זרם)



השוואת זרם - דגימת זרם (זרם, זרם)



$$D = 1 + \beta A'_v, \quad A_{vf} = \frac{A'_v}{D}, \quad R_{if} = R_{in} D, \quad R_{of} = R_o D$$

$$D = 1 + \beta A'_i, \quad A_{if} = \frac{A'_i}{D}, \quad R_{if} = \frac{R_{in}}{D}, \quad R_{of} = R_o D$$

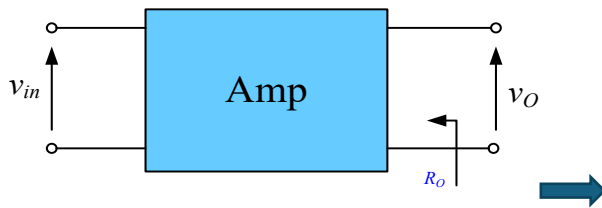
משוב חיובי: קריטריון ברקהאוזן - Barkhausen Criterion

$L(s)$ - Loop Gain: $L(j\omega) = A(j\omega)\beta(j\omega)$

$|L(j\omega)| = 1, \quad \angle L(j\omega) = 0 \pm 2\pi k, \quad k = 1, 2, 3, \dots$

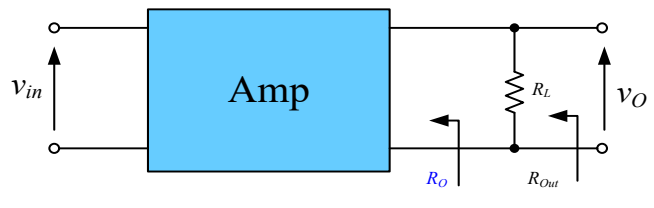
$A(j\omega)\beta(j\omega) = 1 \angle 0 \pm 2\pi k$

$|A(j\omega)\beta(j\omega)| = 1, \quad \angle A(j\omega)\beta(j\omega) = 0 \pm 2\pi k$



$$A_{VO} = \frac{v_O}{v_{in}} \Big|_{R_L \rightarrow \infty} = G_m R_o$$

$$R_o \Big|_{R_L \rightarrow \infty}$$



$$G_v = A_v = \frac{v_O}{v_{in}} \Big|_{R_{L_Final}} = A_{VO} \parallel R_L = G_m R_o \parallel R_L$$

$$R_{out} \Big|_{R_{L_Final}} = R_o \parallel R_L$$